Yuchen Wang

Ph.D. Student in Computer Science

Department of Computer Science, William & Mary

Phone: (1) 757-332-8055 • Email: ywang142@wm.edu • Homepage • LinkedIn

Education

William & Mary Sep. 2024–Present
Ph.D. in Computer Science Department of Computer Science

Shanghai Jiao Tong University Sep. 2021–Mar. 2024
M.S. in Control Science & Engineering School of Electronic Information and Electrical Engineering

North China Electric Power University Sep. 2017–Jul. 2021
B.E. in Automation School of Control and Computer Engineering

Research Areas

Physics-informed machine learning; world models; world model-based reinforcement learning.

Publications

[1] **Yuchen Wang***, Jiangtao Kong*, Sizhe Wei, Xiaochang Li, Haohong Lin, Hongjue Zhao, Tianyi Zhou, Lu Gan, and Huajie Shao. WestWorld: A Knowledge-Encoded Scalable Trajectory World Model for Diverse Robotic Systems. *International Conference on Machine Learning (ICML)*, 2026. **Spotlight** (2.2%).

[2] Xiaochang Li, Chen Qian, Qineng Wang, Jiangtao Kong, **Yuchen Wang**, Ziyu Yao, Bo Ji, Long Cheng, Gang Zhou, and Huajie Shao. Lens: A Knowledge-Guided Foundation Model for Network Traffic. *Transactions on Machine Learning Research (TMLR)*, 2026.

[3] Zhenyu Zong, **Yuchen Wang**, Haohong Lin, Lu Gan, and Huajie Shao. A Generalizable Physics-guided Causal Model for Trajectory Prediction in Autonomous Driving. *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[4] **Yuchen Wang**, Hongjue Zhao, Haohong Lin, Enze Xu, Lifang He, and Huajie Shao. A Generalizable Physics-Enhanced State Space Model for Long-Term Dynamics Forecasting in Complex Environments. *International Conference on Machine Learning (ICML)*, 2025.

[5] Hongjue Zhao, **Yuchen Wang**, Hairong Qi, Zijie Huang, Han Zhao, Lui Sha, and Huajie Shao. Accelerating Neural ODEs: A Variational Formulation-Based Approach. *International Conference on Learning Representations (ICLR)*, 2025.

[6] Hongjue Zhao, **Yuchen Wang**, Hairong Qi, Jiajia Li, Lui Sha, Han Zhao, and Huajie Shao. Accelerating Neural Differential Equations for Irregularly-Sampled Dynamical Systems Using Variational Formulation. *ICLR Workshop on AI4DifferentialEquations in Science*, 2024.

[7] **Yuchen Wang**, Can Xiong, Changjiang Ju, Genke Yang, Yu-wang Chen, and Xiaotian Yu. A Deep Transfer Operator Learning Method for Temperature Field Reconstruction in a Lithium-Ion Battery Pack. *IEEE Transactions on Industrial Informatics*, 2024. (IF=**11.7**).

[8] **Yuchen Wang**, Can Xiong, Yiming Wang, Po Xu, Changjiang Ju, Jianghao Shi, Genke Yang, and Jian Chu. Temperature State Prediction for Lithium-Ion Batteries Based on Improved Physics-Informed Neural Networks. *Journal of Energy Storage*, 2023. (IF=**8.9**).

Preprints and Submissions

[1] Hongjue Zhao, Yizhuo Chen, **Yuchen Wang**, Hairong Qi, Lui Sha, Tarek Abdelzaher, and Huajie Shao. Understanding the Theoretical Foundations of Deep Neural Networks through Differential Equations. *arXiv preprint*, 2026.

Research Experience

Research Assistant Sep. 2024–Present
William & Mary Advisor: Prof. Huajie Shao

- First-authored *WestWorld: A Knowledge-Encoded Scalable Trajectory World Model for Diverse Robotic Systems*, accepted to ICML 2026 as a Spotlight paper.

- First-authored *A Generalizable Physics-Enhanced State Space Model for Long-Term Dynamics Forecasting in Complex Environments*, accepted to ICML 2025.
- Co-authored *Lens: A Knowledge-Guided Foundation Model for Network Traffic*, accepted to TMLR 2026.
- Co-authored *A Generalizable Physics-guided Causal Model for Trajectory Prediction in Autonomous Driving*, accepted to ICRA 2026.
- Co-authored *Accelerating Neural ODEs: A Variational Formulation-Based Approach*, accepted to ICLR 2025.
- Studied generalizable dynamics modeling, physics-guided representation learning, neural differential equations, and robotic world models.

Research Intern Nov. 2023–Jun. 2024
William & Mary Advisor: Prof. Huajie Shao

- Co-authored *Accelerating Neural Differential Equations for Irregularly-Sampled Dynamical Systems Using Variational Formulation*, accepted to the ICLR 2024 Workshop on AI4DifferentialEquations in Science.

Research Assistant Sep. 2021–Jan. 2024
Shanghai Jiao Tong University Advisors: Prof. Changjiang Ju and Prof. Genke Yang

- First-authored *Temperature State Prediction for Lithium-Ion Batteries Based on Improved Physics-Informed Neural Networks*, published in *Journal of Energy Storage*.
- First-authored *A Deep Transfer Operator Learning Method for Temperature Field Reconstruction in a Lithium-Ion Battery Pack*, published in *IEEE Transactions on Industrial Informatics*.
- Designed a production-planning sub-algorithm for the National Key Research and Development Program of China project on factory operating systems for intelligent manufacturing in process industries.
- Developed algorithms for lithium-ion battery state of health, temperature, and safety estimation in a Ningbo City-funded battery systems project.
- Completed graduate thesis: *Research on Key Technologies for Temperature Field Reconstruction in a Lithium-Ion Battery Pack Based on Physics-Informed Machine Learning*.

Research Assistant Mar. 2020–Mar. 2021
North China Electric Power University Advisor: Prof. Tongxiang He

- Applied PID control in Single-Input-Single-Output and Cascade Three-Input-Three-Output setups for boiler water-level and feedwater-flow management.
- Developed Distributed Control System configuration diagrams using PAS-300 software.
- Completed undergraduate thesis: *A Study on Superheated Steam Temperature System Using the State Feedback Control with Error Integral Based on Simulink*.

Work Experience

Algorithm Engineer, Battery Management System Jun. 2022–Jul. 2023
Research and development role Advisor: Dr. Can Xiong

- Led development of an LSTM-based algorithm for lithium-ion battery temperature-state estimation.
- Contributed to an Extended Kalman Filter-based algorithm for lithium-ion battery state-of-charge estimation.
- Participated in a Gaussian Process Regression-based algorithm for lithium-ion battery state-of-health prediction.

Intern Jun. 2021–Sep. 2021
Ningbo Industrial Internet Institute Advisor: Prof. Changjiang Ju

- Developed industrial PLC HMI software in C#, supporting firmware updates and file operations.
- Participated in the preparation of a research report on the hydrogen energy industry.

Honors and Awards

- ICML Gold Reviewer Award (top 25%), 2026
- Outstanding Graduate of Shanghai Jiao Tong University, 2024
- Graduate Student Merit Scholarship, 2023
- Shanghai Jiao Tong University Second Academic Scholarship, 2022
- College-Level Third-Class Scholarship, North China Electric Power University, 2020
- Alumni Scholarship, 2019